Loop Gain: what a feed, a symptom and a plate of sand have in common

A week of doing our own research, with the failures left in

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Why this document exists

This is a long-form account of one week of research in a small public repository, written to be read aloud and listened to. It is for people who are early in research, or outside institutions, and who have heard the phrase "do your own research" used as a sneer. We want to show what the phrase can mean when it is taken seriously: a question, a model you can run on a laptop, a prediction written down before the run, a collaborator who reads your work against you, a theorem prover that will not accept a claim you cannot prove, and a record that keeps the failures at the same length as the successes.

The week produced a three-minute audiovisual piece called *Loop Gain*, built for a programme called Phase Transition. It also produced two simulation benches, twenty-one machine-checked theorems, several pre-registered hypotheses of which two failed in instructive ways, and one idea that we did not have on Monday: that two things we had been treating as separate were one mechanism at two settings. This document walks through all of it in order. Nothing in it requires mathematics beyond ratios, and every number quoted is in the repository with the code that produced it.

A word on how it was made. One of us, Jeff, works with an AI assistant, Claude, which wrote most of the code and most of the first drafts, including this one. The other, Alexandra, who goes by Lex, works with a different assistant on purpose, so that the two of us do not share one set of blind spots. The repository has a standing rule that nothing is evaluated by the process that produced it. We have tried to honour that rule and we say where we could not.

Part one: the premise

Two briefs that looked unrelated

The repository holds a series of concept briefs for interactive pieces. Two of them matter here.

The first is called *The Unframeable Now*. It is about how events reach us already framed: as cards in a feed, with timestamps, source labels, a sentiment colour, a "related" link. The brief proposed suspending that framing with operators borrowed from the twentieth-century avant-garde, cut, expose the apparatus, collide scales, refuse duration, and watching what remained. Its central claim

was a single sentence: *every frame applied re-enters as signal.* Whatever you do to the feed, your doing it becomes one more thing in the feed.

The second is called *The Return of the Excluded*. It is about pollution, read not as inert waste but as communication. In the same way a symptom carries information about a one-sided attitude, the brief argued, environmental or systemic pollution carries information about patterns of awareness that have been split off. Its central claim was also one sentence: *excluding the signal repeats the act that produced it.* Pushing the symptom out of awareness is the same gesture that caused the dissonance, and it supplies fresh material for the next return.

Lex extended the second brief with a reading from predictive processing and the neuroscience of allostasis: exclusion is not free. Each act of suppression spends something from the body's budget, and the exterior where the excluded material accumulates is also somatic memory. She proposed tracking an allostatic load alongside the exclusions.

These briefs were written as art. We built one of them as a working piece first, the feed, and it turned into a research instrument almost by accident, because every change in it had to be traced to a cause and every cause had to be stated as a rule. Once the rules were written down, they could be run, swept, and proved against.

The question we put to both

A programme called Phase Transition asked teams to translate a research finding into an art form, under the rule *carry the structure, not just the essence.* Our prompt was: what does sound look like, and how can it be organised in an image? That prompt leads, historically, to a plate of sand. In 1787 Ernst Chladni bowed a metal plate strewn with sand and watched the grains leave the places that vibrated and gather on the lines that stayed still. Each tone drew its own figure. Between tones nothing happened. At a tone, a new figure.

The structure under that observation is the one this whole week turned on: a continuous drive meets a threshold, and the pattern reorganises rather than slides. Physicists call the general case a phase transition. Water does not get gradually more solid as it cools; at zero degrees it becomes ice. A programme named for that idea had, in effect, asked us whether our two briefs had a threshold in them.

So we asked. As the feed is turned up, does the reader's end state slide or drop? As the exclusion loop is turned up, does the pollution slide or drop? And if the answers differed, why?

Part two: methods

Benches, not demos

Both pieces were re-stated as headless models: code that runs the same rules as the interactive page but without a screen or a clock, so that it can run a three-minute session in a few milliseconds and run thousands of them. We call these benches. A bench is a place to run the model as a simulator across personas, settings and random seeds, and to compare the results, without touching the piece itself.

The feed's bench carries a conformance test: it runs a scripted session on the real page, headless, on the real clock, and in the bench on the simulated one, and compares the end states. If the bench

and the page disagree, the page is right and the bench is wrong. That sentence is in the bench's README and it is the rule that keeps a bench honest.

Seeds

Every run is seeded. A seed is a number that fixes the random choices, so that the same seed gives the same session every time, on any machine. This matters for two reasons. First, a result can be reproduced exactly. Second, when a run looks strange, you can run that one again and look at it, rather than hoping it recurs. When we say "twenty seeds a point" we mean twenty different sessions at each setting, each one repeatable.

Sweeps and a verdict rule

A sweep pushes one constant of the model through a range and records the end state at each point. We used twenty-four points, spaced evenly on a log scale where the quantity was a rate and linearly where it was a probability, and ten or twenty seeds a point. The end state we read was the median over seeds.

Before the first sweep we wrote a rule for reading the curve, so that we would not be deciding afterwards what counted as a drop. A curve that moved less than a stated amount was *flat*. One where a single step between neighbouring points carried a third or more of the whole fall was a *cliff*, because that is the shape a discontinuity has at our grid spacing. One with no such step but a fall concentrated in a narrow band was *steep*. Everything else *slid*. The rule is in the sweep script, and the README records that it was wrong twice, in both directions, and why it was kept.

Pre-registration

For the second bench we wrote the hypotheses to disk before any run. Each said what we expected, with a number, and what would falsify it. The section of the README holding them was marked so that nothing above the line would change after the runs, and nothing did. When a hypothesis failed, the failure stayed, with its analysis, at the same length as a success would have had.

This is the single most useful discipline we can recommend to anyone doing their own research. It costs ten minutes. It makes the difference between a result and a story.

Proofs

Where a claim followed from the rules alone, with no numbers needed, we proved it. We used Lean 4, a proof assistant, in its core form with no mathematical library, so that the proofs could be checked with one small toolchain and read without a library behind them. A proof assistant does not accept a claim because it is plausible; it accepts a claim when every step is justified from the definitions. The rules of each model were transcribed by hand into Lean, levels as whole numbers in thousandths, time in milliseconds, and the theorems were stated in the form "if the model is as written, then this follows." None of them is a claim about a person.

The transcription is the weak point, and we say so. If the model changes, the transcription has to be checked, the way the conformance test checks the bench against the page.

Reading the literature

Where the models touched claims about people, we went to the published literature, from abstracts, and flagged every number as something to check before citing. We did not read the full texts of most papers, and we say so where we quote them.

The rule we could not fully keep

The repository's standing rule is that nothing is evaluated by the process that produced it. In this week, one process, Claude working with Jeff, wrote the briefs into rules, chose the constants, wrote the hypotheses, ran them, and drafted the write-ups. The pre-registration is what stood between the hypotheses and the results. Lex read each result against the translation test she had set. The proof assistant checked the theorems. Those are three outside checks, and they are not the same as an independent replication. We would welcome one.

Part three: the simulations

The feed

The feed's model is small enough to describe in full. A stream of items arrives from seven channels, four that ask something of the reader (social, news, work, family) and three she would rather be looking at (garden, art, music). The asks arrive about three times as often, and the ranking favours them, by stated weights. Every gesture she makes, a scroll, a tap, a like, a hide, a mute, a refresh, returns to the stream 2.2 seconds later as one more item. A band shows the share of the stream that is her own echo.

Four dimensions wear down under what the session records: attention, which falls with arrivals and taps and recovers only after three seconds of quiet; calm, which follows a pulse the feed sets; agency, which falls with the share of the stream that is her echo and recovers only when a like on something she wanted actually worked, or when she uses the one permanent act, the Empty Bin; and the minutes she meant for something else, which drain with the clock. Lex's modules added a moment on a bad day that strikes when her hands are still, an allostatic floor that rises with each moment and never recovers, slow asymmetric recovery of the pulse, and a fourteen-second refractory window.

Eight personas use the feed in eight ways: Kate, who does her seven things and then scrolls now and then; Kate who stops; a doomscroller who never stops; a tidier who hides and mutes everything that asks; a gardener who likes only what she wanted; an abstainer who opens it and does nothing; and two who use the Bin, one decisively and one who cannot choose.

What the sweep found

We swept four constants: the interval between arrivals, from twelve seconds down to a third of a second; the asks' weight against the wants', from an eighth to sixteen times; the time a move takes to return; and the time an item lasts. We ran the arrival interval on a normal day and on a bad day with the moment.

No cliff anywhere. Every curve was flat or slid. What there was instead was a knee: on the arrival interval, every persona held flat from twelve seconds down to about three and a half seconds between arrivals, which is the page's normal day, and then slid over roughly a factor of ten. The tidier fell furthest, from 0.64 to 0.07 on a wholeness scale from zero to one. The bad day shifted every curve down without changing its shape.

Where the knee came from

We read the knee off the rules by hand. Attention recovers only on a tick when nothing has arrived for three seconds. The gap between arrivals is the interval times a random factor between 0.6 and 1.5. So the share of gaps long enough for a recovery is a simple function of the interval: it is one when the interval is above five seconds, zero when it is below two, and a smooth curve between. That function is continuous. Across the twenty-four points of the sweep, each persona's attention at the end correlated with it at between 0.90 and 0.997. The knee was a threshold in the rules meeting a rate, not a reorganisation of the state. A slope, with a reason.

What was proved about the feed

Four claims turned out to be theorems of the rules, not findings of the sample. The band cannot be kept flat by acting: the Bin severs at most one branch per session and using it costs two moves, so any session with two or more moves has a live echo at some moment. No rule lowers the allostatic floor, and none takes the pulse under it. A moment lowers the ceiling calm relaxes toward by exactly its step, and the ceiling never rises again in the session. Without a like that worked or an emptying, agency never rises; echoes ageing out is not among the rules that move it, so being buried is not agency. All four check in Lean with no gaps.

One of those proofs corrected us. The draft of the floor theorem was false as first written: if the floor were ever above its cap, the moment's capping rule would pull it back down. The theorem needed the hypothesis that the floor starts under the cap, which the page's invariant guarantees and the draft had overlooked. The compiler found it. That is what a proof assistant is for.

The exclusion loop

The second bench was built from Lex's extended brief. A field of awareness. Signals appear in it at a steady rate: the system's own information. The viewer can exclude a signal, which cleans the field and deposits the signal's mass in a shared exterior, or hold it, in which case it dissolves after six seconds of attention. Above a threshold of accumulated mass, the exterior returns signals to the field, and a returned signal carries a gain: it is louder than what left. Each exclusion raises an allostatic load that recovers slowly. The load's effect was kept as a frame, because the brief did not decide it: in the *fatigue* frame the load lowers the capacity to exclude; in the *reactivity* frame it raises it; in the *none* frame it does nothing. A fourth frame came later.

The quantity the brief did not name

Before running anything, we wrote down what the loop is. Each exclusion deposits a unit of mass. A unit of exterior mass either returns, at some rate, or dissipates, at another, so it returns with a

probability equal to the return rate over the sum of both rates. A returned signal is excluded with some propensity and, if it is, deposits the gain times a unit. So one exclusion begets, on average,

propensity × gain × return rate ÷ (return rate + dissipation rate)

exclusions in the next generation. Epidemiologists call this number R0, pronounced “R nought”: the number of new cases one case causes. A branching process with R0 below one dies out. At one it is critical. Above one it multiplies. The brief’s claim that ignoring the signal supplies fresh material for the next return is the claim that the gain is above one, and with gain above one there is a propensity to exclude above which the viewer cannot keep up with what she causes.

With the bench’s constants, the return share came to 0.83 and the gain to 1.5, so R0 equalled 1.25 times the propensity, and the critical propensity was 0.8. We wrote that down as the prediction.

Five hypotheses, written first

We pre-registered five. The loop has a critical point near 0.8 and the sweep will read it as a cliff or steep. Without gain there is no critical point. Fatigue bounds the pollution and the body pays to its cap; reactivity accelerates and moves the critical point left. Destroying the excluded content and returning only the act moves the threshold without removing it. And a viewer who never excludes ends with an empty exterior.

What happened

The first hypothesis failed under its own test. The exterior at the end rose smoothly from zero to 128 across the range of propensity, and the verdict rule said “slides.” The sharpest step was at the top of the range, not at 0.8. We had written a rule for a bounded quantity that steps; a quantity that diverges at a critical point does not step, it grows, and in a three-minute session it grows smoothly through the critical value. The pre-registered instrument was the wrong one for the prediction it was testing. We recorded that as the failure it is.

Then we wrote a different test, labelled post-hoc so no one could mistake it for the pre-registered one. For a critical point, the right question is whether the end state depends on how long the session ran. Below the critical point it should settle: a twelve-minute session ends about where a three-minute one did. Above it should keep growing. At a propensity of 0.6, the twelve-minute exterior was 1.1 times the three-minute one. At 0.8, 3.9 times. At 0.9, 62 times. At 1.0, 183 times. The turn was at the predicted 0.8. The mean-field number described the simulator.

The second hypothesis held: with gain of one, nothing diverged at any propensity. The third held in half: fatigue bounded the exterior under six at every setting, but the body did not pay to its cap; it settled near half, because a body that excludes less also loads less. Reactivity moved the critical point from 0.8 to about 0.4. The fourth was consistent by magnitude and became a theorem. The fifth held trivially.

The fourth frame

Daniel Wegner’s ironic process theory, from 1994, explains why trying not to think of something makes you think of it. Two processes run at once: an operating process that does the suppressing,

effortful and conscious, and a monitoring process that scans for the suppressed thought, cheap and unconscious, and in scanning keeps the thought accessible. Under load, stress, fatigue, time pressure, the operator weakens and the monitor keeps running, so the suppressed content becomes more accessible, not less. The rebound is real and small; a 2001 meta-analysis by Abramowitz, Tolin and Street put it at a small-to-moderate effect.

Wegner's claim differed from both of Lex's frames. Load does not change how readily you exclude; it raises the rate at which the excluded returns. We added that as a fourth frame, *ironic*, in which the return rate scales with the load, and we wrote the prediction first: the critical point would move only a little, from 0.80 to no lower than 0.73, because the return share was already 0.83 and doubling the return rate could only buy the remaining dissipation share; and the frame would be a fraction of reactivity in size.

Half right. The critical point moved exactly as far as predicted and no further. The size claim was wrong by an order of magnitude. At full propensity the exterior ended at 1107 against 128 for the brief as written, with 3148 exclusions against 385. The mean field had missed it because R0 counts exclusions per generation and says nothing about how long a generation takes. Doubling the return rate barely moves R0 and halves the time a generation takes, so above the threshold the exponential growth rate roughly doubles.

This is the sentence we did not have on Monday: **R0 says whether the loop runs away; the return rate says how fast.** Wegner's monitor does not change whether suppression fails. It changes how quickly the failure compounds.

What was proved about the loop

Ten theorems, checked. With R0 at or above one, the excluded mass after n generations is at least the starting mass plus n times the new material: unbounded while the system keeps speaking. With R0 below one, there is a bound the mass never passes once under it. With exclusion the only act and no dissipation, the exterior never decreases. The load never falls under the viewer's acts alone. Destroying content and returning only the act does not raise R0. A higher return rate never lowers R0. And R0 never exceeds propensity times gain, whatever the return rate does, which is exactly the correct half of the fourth frame's prediction and is silent about the speed. That division, between what a theorem about a ratio can say and what needs a simulator with a clock, is itself a finding.

Part four: the findings, together

One loop at two gains

The feed's loop and the exclusion loop are the same mechanism at two settings. In the feed, a gesture returns once, as itself, and the echo does not act. Its reproduction number is zero. The echo share accumulates linearly and ages out. The sweep found a slope, and the slope had a reason. In the exclusion loop, a return invites the act that produced it and comes back louder. Its reproduction number is positive, and a positive reproduction number has a critical point. The sweep found a cliff, where the number said it would be.

Every frame applied re-enters as signal and *excluding the signal repeats the act* are the same sentence at R0 equals zero and R0 greater than zero. The difference between two briefs that had seemed to belong to different disciplines, media theory and depth psychology, is one rule: does what returns invite an act?

Whether and how fast

The reproduction number decides whether a loop runs away. The return rate decides how fast. These are different quantities, and the mathematics of one is silent about the other. We proved the first and could only simulate the second. Anyone building a model of a feedback process, in any field, should expect both and should not read either as the other.

The small gain in the literature

The human rebound effect is small: around d = 0.3 by the secondary sources we reached. In the bench's terms that is a gain a little above one. With a return share of 0.83, a gain under 1.2 has no critical point at any propensity; a gain of 1.5 has one at 0.8. Whether a person's loop has a gain, and what it is, is not a question a bench can answer. It is a question for a person and a measurement, and we say so everywhere the numbers appear.

What the sand already knew

Chladni's plate is the same structure in its oldest form. Sweep the tone continuously and the sand does nothing between resonances, then snaps into a new figure at each one. A continuous drive, a threshold, a reorganisation. Our piece puts the three readings on one signal, in order: the feed's slope, the loop's threshold, the plate's form.

Part five: the piece

Loop Gain is three minutes, recorded from a browser, built in one day from the week's parts.

In the first movement, the feed's model runs headless and is replayed with its sound: a two-tone ping for what asks, a soft fall for what she would rather, a click for her move and a pop when it comes back. The line draws the stream by ranking, her moves above it with an arc to each return, and the band beneath. Loop gain zero.

In the second movement, the same pings are fed into a delay of less than half a second with a gain that rises from 0.8 to 1.25 over about a minute. Below one, the echoes thin and die. Past one, every return begets more than itself and the loop blooms into a howl: the Larsen effect, which is what happens when a microphone hears its own speaker. The chart on screen is the loop's measured level, from an analyser on the audio itself, against the gain ramp, with R0 equals one marked where the gain crosses one. Two things are placed by hand and said on screen: a resonance at 870 hertz, as a room has one, and a soft clip where a speaker's ceiling would be.

In the third movement, the howl's settled frequency, read from the loop's spectrum, drives the plate. The sand gathers into the figure that tone draws. The runaway grounds into form.

Two of Lex's three perceptual favourites from the review are in the piece, in roles that carry the research rather than decorate it. Bridson's Poisson-disc growth draws the exterior as a disc that fills by what the loop carries and regrows in a new colour where the gain crosses one. Atlas keeps the piece's own map in the corner: the three movements as a ring, the needle at the playhead, growth planted where each event was heard.

What is honest about it: the line replays a real run of the model; the level is measured, not modelled; the plate's tone is read, not typed. What is not: the movements are cut by the clock, not by the sound, and the body, Lex's load, is left out, because three minutes cannot hold a fifth variable and say what it did.

Part six: doing your own research

If you are early in this, here is what the week suggests, in the order it mattered.

Write the rules down. A piece, a claim, a hunch becomes research the moment its rules are stated plainly enough to run. Most of our findings came from noticing something in a rule we had already written.

Seed everything. A result you cannot reproduce exactly is a story.

Predict before you run. Write the number down. Write what would falsify it. Mark the section and do not touch it after. Our two most useful results this week were a hypothesis that failed because its test was wrong, and a hypothesis that was half wrong by an order of magnitude. Neither would have taught us anything if we had written them afterwards.

Keep the failures at full length. The verdict rule that was wrong twice is still in the repository, unchanged, with a note each time. A reader can see the instrument being corrected. That is worth more than a clean record.

Prove what can be proved. A proof assistant is free, runs on a laptop, and will find the hypothesis you forgot. Ours found one. It also tells you, by what it cannot prove, where the simulator's work begins.

Find the quantity. Most feedback systems have a number that decides them. The brief about pollution had been written, read and extended by two people before anyone wrote down its reproduction number. Once it was written down, the week organised itself around it.

Work with someone who reads against you. Lex set the translation test, read every result, chose a different assistant so we would not share blind spots, and asked to be credited as a reviewer rather than an author because that is what she did. Both choices made the work better and the record truer.

Say what the numbers are not. Every constant here is a choice. Every theorem is about the model. No sentence in this document is a claim about what a feed, or suppression, or pollution does to a person. The literature that bears on that is cited, from abstracts, flagged to check. The opportunity of doing your own research is real; so is the obligation to say what you did and did not do.

What is left

A replication by someone who did not build any of this. A person, not a persona, playing a bad day with a heart-rate reading, which is the only way the feed's model becomes evidence about a body. The reverse arrow, a soundtrack that lags the pulse and leads it down, which the literature review found no one has run. And the exclusion loop's gain, measured rather than chosen.

References

Pointers, not a bibliography. Each was reached through abstracts or secondary sources this week and should be checked against the original before being cited.

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